APPLICATION OF AN EXTENDED TRAVELLING FIRE METHOD (ETFM) FRAMEWORK ON THE VESELÍ TRAVELLING FIRE TEST BUILDING USING OPENSEES

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Abstract

Travelling fire is a type of design fire load for structures with large compartments now being adopted for performance-based structural design. An extended travelling fire method (ETFM) framework has been developed by the authors, based on a mobilized version of Hasemi's localized fire model, which quantifies the impact of a fire on adjacent structural members, and a smoke layer calculation, using the FIRM zone model, which considers mass and energy balance in the compartment, implemented in OpenSees. Performance-based design for structures in fire requires validated methodologies, here for the first time a travelling fire design method is demonstrated via direct comparison against experimental measurements, those obtained in the Veselí travelling fire (a two-storey steel-composite structure). Despite the over-simplification of the model in treating the fire spread process as a constant rate, a fair agreement is achieved in the thermal response of the structural elements, though there are significant under-predictions during the decay phase of the fire.

Keywords: travelling fires; performance-based design; structural fire engineering; large compartments; OpenSees fire module

1. Introduction

In structural fire design, a key principle is to ensure that the fire resistance of a structure is greater than the fire severity. However, this principle would be undermined and difficult to assess if inappropriate fire exposure models are adopted for the design compartment. The problem for structural engineers is particularly challenging when design compartments are very large and conventional fire exposure models are inappropriate. This kind of problem is now being tackled with 'travelling fire' methodologies [1], which are conceptualised as fires that may burn locally and which will tend to move across entire floor plates over a period of time in large compartments. There are three main representations of travelling fires which can be found in the literature: Clifton's model [2], Rein's model [3], and an extended travelling fire methodology (ETFM) framework *conceptually* put forward by the authors in 2016 [4, 5]. The ETFM framework is developed by 'mobilising' Hasemi's localized fire model [6] for the fire

plume near the structure (i.e. near-field), and combined with a simple smoke layer calculation utilising the FIRM zone model [7] for the areas of the compartment away from the fire (i.e. far-field). The heat fluxes generated by the ETFM framework will enable both a heating phase and a cooling phase for each structural member in the large compartment, as shown in the Fig. 1.

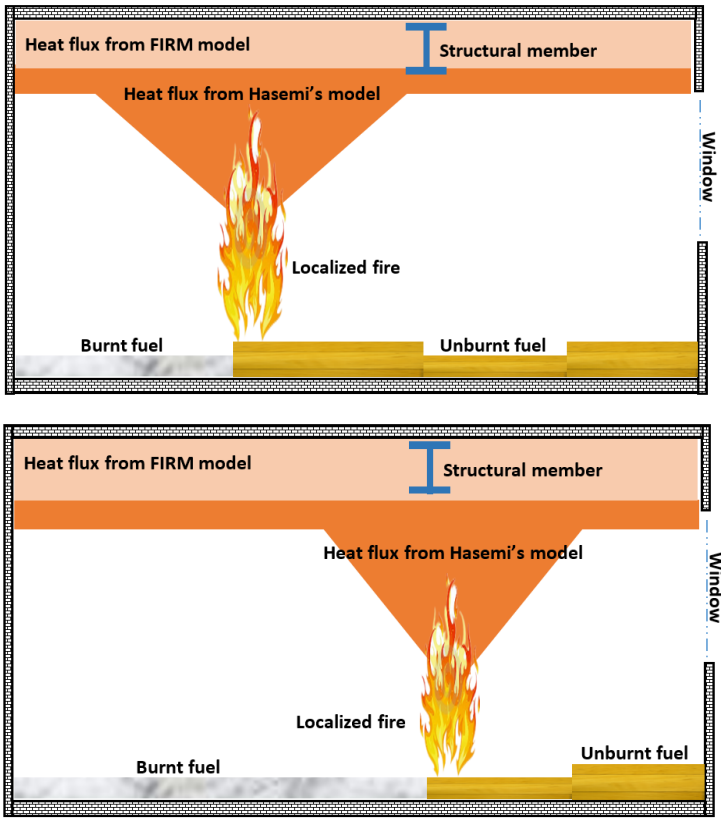

Figure 1. Heat flux at time t and t+Δt respectively

The ETFM framework enables the analysis to capture both spatial and temporal changes of the thermal field. Fire temperatures are variable for the near field, contrasting the uniform 800°C-1200°C assumption in Rein's model, while all elements in one firecell share the same fire exposure history in Clifton's model. Furthermore, adoption of the FIRM zone model also enables the ETFM to consider smoke accumulation under the ceiling, which is ignored in both previous models. Crucially, by utilising the FIRM zone model into the ETFM framework energy and mass conservation are both enforced for the design compartment. By contrast previously proposed travelling fire methodologies simply force other existing models to 'travel' (i.e. modified parametric fire curves in Clifton's model, 800°C-1200°C temperature block with Alpert's ceiling jet model in Rein's model), and have not attempted to explicitly account for the mass and energy balance in the compartment, thus the ETFM framework has a more rigorous basis than previous models. The work presented here is a *demonstration* of this performance-based design approach for the structure within a large compartment with the travelling fire model implemented in OpenSees, i.e. the ETFM framework.

Performance-based design for structures in fire requires validated methodologies. It is apparent that none of the travelling fire design methods mentioned above have been directly compared against any travelling fire experiments so far. This paper demonstrates the application of ETFM framework to representation of fire spread in a real building - the Veselí travelling fire test

building (a two-storey steel-composite structure [8]). It is achieved through performing a numerical investigation of the thermal response of the structural elements, i.e. comparing the cross-sectional time-temperature evolutions of the instrumented steel beams in the test compartment, between the predicted time-temperature histories using the ETFM framework and the test data. All these analyses are performed using the SIFBuilder [5], which is an OpenSees-based integrated computational tool for performing automated thermo-mechanical analyses for large structures under realistic fires.

2. Implementation of the ETFM framework in OpenSees

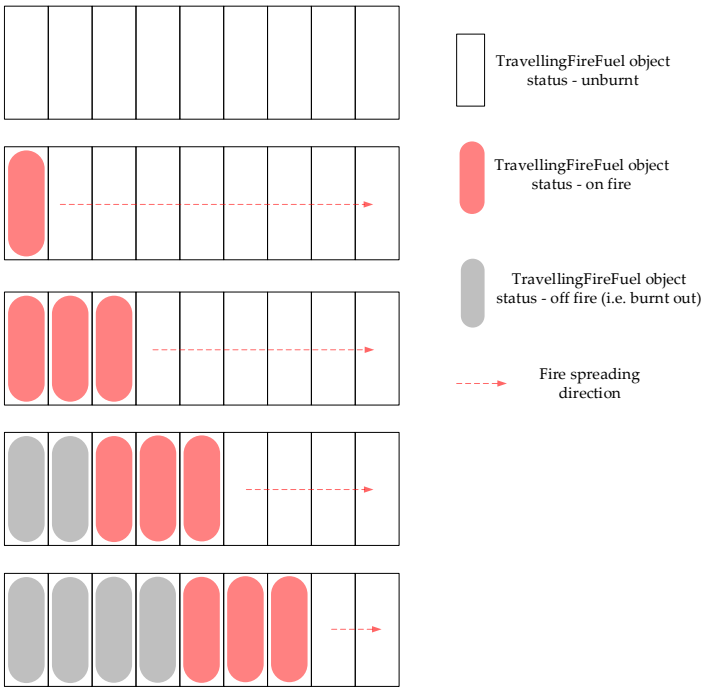

Figure 2. A slice of burning fuel bed meshed with *TravellingFireFuel* objects when the fire spreads.

The ETFM framework is based on a 'mobile version' of Hasemi's localised fire model combined with a simple smoke layer calculation – the FIRM zone model. The *ETFM_MovingHasemi* class and *ETFM_ZoneModel_FIRM* class are added to the OpenSees fire module, to calculate the spatially and temporally non-uniform heat fluxes for different structural elements, produced from the 'summation' of the heat fluxes from the Hasemi localised fire model and the FIRM zone model (see **Error! Reference source not found.**), r espectively. The *ETFM_MovingHasemi* class is adapted from the *LocalizedFireEC1* class, which is the original class for the Hasemi's localised fire model implementation in OpenSees developed by Jiang [9]. The 'travelling fire' interacts with the heat transfer module through their respective interfaces at each time step in order to determine the transient fire-imposed boundary conditions to the adjacent structural surfaces (see **Error! Reference source not f ound.**, only one structural element shown for clarity). In addition, the *ETFM_ZoneModel_FIRM* class is for simulation of the transient generation of a hot smoke layer at the compartment ceiling, with natural ventilations, through considering both the energy and mass conservations.

In order to transiently determine the burning fuel status (e.g. transient fire location and fire size) as the fire is evolving in the large compartment, a *TravellingFireFuel* class is introduced in the OpenSees fire module. **Error! Reference source not found.** schematically demonstrates a slice o f burning fuel bed meshed with *TravellingFireFuel* objects when the fire spreads. *TravellingFireFuel* is used to account for the distributed meshed fuel cells, and each instantiated *TravellingFireFuel* object has its own attributes. These attributes include the coordinates of the fuel cell in the compartment, size of the fuel cell, fire spread rate on the fuel cell, fuel load density, maximum HRR per unit area, and burning status (i.e. unburnt, burning, and burnt out), etc. This feature relies on the object-oriented nature of OpenSees using C++, enabling each fuel cell (i.e. *TravellingFireFuel* object) to have its own transient status and identities. It further complies with the nature of the ETFM, as a framework. For instance, once a fire spread model is developed for the ETFM framework in the future, it can be easily associated with the fire spread rate variable for each *TravellingFireFuel* object, to 'propagate' the fire accordingly based on each fuel cell's identities (e.g. fuel load type, the relative location of the fuel cell and ventilation openings, etc.). Moreover, the *TravellingFireFuel* class also facilitates convenient implementation of the lumped fuel in the ETFM framework, as each fuel cell is 'consumed' independently. A *TravellingFireFuel_Iter* class is introduced for looping over all the meshed fuel cells (i.e. *TravellingFireFuel* objects) to check if the fuel is currently on fire, or not, at each time step, then aggregating them as the entire burning fuel area. Along with the spatially and temporally changed heat fluxes induced by the ETFM framework, heat transfer analysis is performed to estimate the time-temperature histories of the structural members. This may include the effects of preheating, direct heating, post-heating and cooling.

3. Veselí Travelling Fire Test

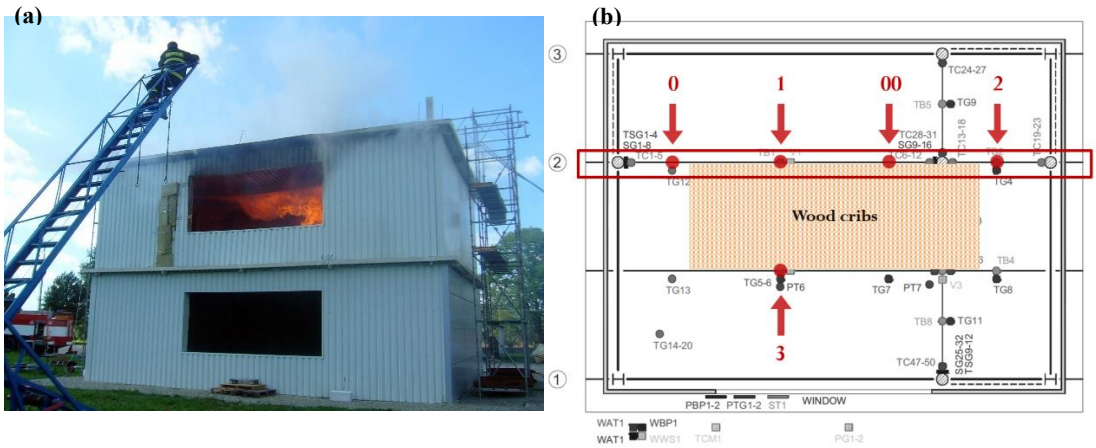

Figure 3. (a) Experimental building during the Veselí Travelling Fire Test [1], and (b) fuel load scheme in hatched 8m × 3m, and investigated beam temperatures at locations 0, 1, 00, and 2

This test was part of an European-funded project called COMPFIRE [10], investigating the behaviour and robustness of the practical beam-to-column connections under travelling fire scenarios. The experimental building was a 10.4m × 13.4m in plane × 9m in height (139m^2 floor area) two-storey steel composite structure, with a 2m × 5m unglazed opening for each floor to provide enough ventilation for a smooth development of the fire (see Figure 3(a)). The height of each floor was 4m. The fuel load was wood cribs with estimated density 173.5 MJ/m^2,

distributed on the second floor, with a 3m × 8m rectangular shape as shown in Figure 3(b). More details of this fire test can be found in [10].

4. ETFM framework for Veselí Travelling Fire Test

```
# FIRE DEFINITION
AddFire -floor 1 -type ETFM_Travelling_Fire -IgnitionLine point1 2 4 3 point2 2 4 6 -fireTravelDirection AntiClockWise;

# MORE FIRE INFO
AddFirePars -floor 1 -type ETFM_Travelling_Fire -combination 1 1 -plumeModel Zukoski -totalHeatLossFraction 0.68 -radiativeHeatLossFraction 0.25 -ventWidth 5 -sillHeight 1.2 -soffitHeight 3.2;

# FUEL LOAD DISTRIBUTION DEFINITION
AddFuel -RMFD 1 -Floor 1 -RefinedDimension point1 2 4 3 point2 2 4 6 point3 10 4 6 point4 10 4 3 -SpreadRate 6.5 -FuelLoadDensity 680 -HRRperArea 700;
```

Figure 4. Corresponding fire part of the model script in Tcl, using the ETFM framework in SIFBuilder for the Veselí Travelling Fire Test.

As demonstrated in Figure 4, to achieve the comparison between the ETFM framework and Veselí Travelling Fire Test, it is worth noting that the selected input parameters for the ETFM framework have three sources: 1) the exact test setup information, including compartment dimension 12m × 9m, fuel load distribution shape 8m × 3m, opening size 5m × 2m, opening location (i.e. sill height 1.2m), and fire ignition location; 2) estimated values based on the relevant fire test literature, including a global and constant "total heat loss fraction", 0.68, deemed appropriate according to the thermal lining information from the test [11], fuel load density 680MJ/m^2 with an assumed chemical heat of combustion of the wood sticks [12] of 18.0 MJ/kg with combustion efficiency [13] 0.8, fire spread rate 6.5 mm/s based on test observations, and maximum HRR per unit area according to Horová *et al.* [12]; 3) empirical values/formulas, e.g. Zukoski's plume model [14], and radiative heat loss fraction 0.25 as recommended in Janssens [7].

Figure 5 demonstrates the comparison through quantifying beam member bottom flange temperatures at various locations along the fire trajectory, between the predicted time-temperature histories using the ETFM framework (model input parameters shown in Figure 4) and the calculated time-temperature histories, which used the relevant adjacent test thermocouple (TC) data to determine the heat transfer boundary condition. It suggests a good agreement between the ETFM framework and the Veselí Travelling Fire Test, in terms of maximum steel temperatures at various locations along the fire trajectory. Since a constant fire spread rate of 6.5 mm/s is used in the ETFM framework, in contrast to the accelerating fire spread of 3 to 8.9 mm/s as observed in the test, a clear 'offset' of results between the framework and the test can be identified. To 'filter out' this offset due to the assumption of a constant fire spread rate, all time-temperature history predictions from the framework were adjusted with a 4 minute time shift, as shown in Figure 5(a).

Moreover, Figure 5 (b), (c), (d), and (e) present the time-shifted comparison using the beam member bottom flange temperatures between the ETFM framework and the Veselí Travelling Fire Test at locations 0, 1, 00 and 2, respectively. These four figures mostly suggest that the ETFM framework can generate good predictions during the heating stage of the travelling fire at various locations along the fire trajectory, except the location 2, which is at the opposite side of the fire ignition location.

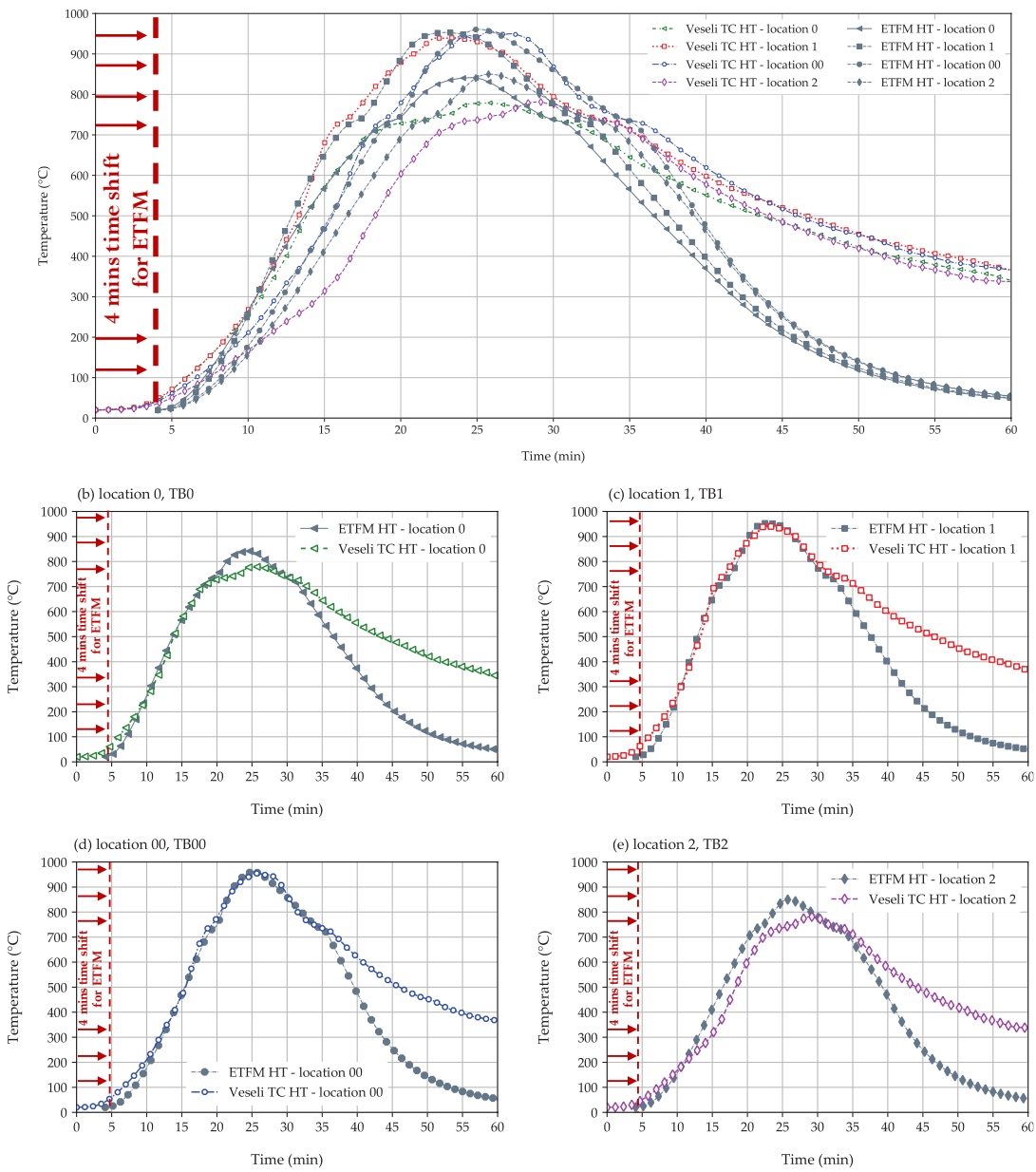

Figure 5. Comparison between the ETFM framework and the Veselí Test (with 4 mins time shift), using steel temperatures at bottom flanges, at various locations in the fire spread direction,

(Veselí thermocouple (TC) temperatures at the vicinity of beam bottom flanges are used as input for SIFBuilder heat transfer (HT)), (a) all locations including 0, 1, 00 and 2, (b) location 0, (c) location 1, (d) location 00, and (e) location 2.)

This suggests that the uniform temperature assumption due to the pre-heating of the smoke upper layer from the FIRM zone model of the ETFM framework may over-predict the temperature at the far-field. However, this overprediction is not of great concern because it generates a more conservative temperature. More importantly, a large discrepancy is found at all four locations during the decaying stage of the travelling fire. This discrepancy is undesirable, as it implies that the predicted time-temperatures from the ETFM framework will

be far lower than the test results, which is probably non-conservative. A possible reason in using the ETFM framework along with SIFBuilder is that as soon as the fire is decaying and even diminishing, the resultant heat release rate (HRR) will decrease drastically. This HRR determines the efficiency of the energy 'pumped up' to the upper smoke layer, which highly affects the resultant design fire temperatures. However, in reality (i.e. during the test) the temperature during the cooling stage is not only affected directly by the fire HRR, but other factors such as the glowing embers and the residual heat in the enclosure boundaries, which also contribute to the resultant thermal exposures in the compartment. It is worth noting that similar discrepancies were also identified by Horová *et al.*[8] through using a CFD model for the Veselí Travelling Fire Test.

5. Conclusions

Application of the ETFM framework to a real building, i.e. the Veselí Travelling Fire Test, representing such a travelling fire scenario, is a means to assess the capabilities and limitations of the framework. The comparison between the ETFM framework with the Veselí Travelling Fire Test confirms the capability of ETFM framework in predicting the maximum steel temperatures, and the important feature of the relative timing of different steel member time-temperature histories along the travelling fire trajectory. In the authors' opinion, these two capabilities are the most important elements for structural fire design under travelling fires. Nevertheless, apart from the inherent limitations of the Hasemi's localised fire model and the FIRM zone model themselves, it is also identified that the uniform fire spread rate assumption, and underestimation of the temperatures during the cooling stage in the ETFM framework, are two important elements requiring to be improved in future work.

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